Modified Checklist for Autism in Toddlers, Revised with Follow-Up

(M-CHAT-R/F)™

Acknowledgement: We thank Joaquin Fuentes, M.D. for his work in developing the flow chart format used in this document.

For more information, please see www.mchatscreen.com
or contact Diana Robins at DianaLRobins@gmail.com

© 2009 Diana Robins, Deborah Fein, & Marianne Barton

Permissions for Use of the M-CHAT-R/F*TM*

The Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F; Robins, Fein, & Barton, 2009) is a 2-stage parent-report screening tool to assess risk for Autism Spectrum Disorder (ASD). The M-CHAT-R/F is available for free download for clinical, research, and educational purposes. Download of the M-CHAT-R/F and related material is authorized from www.mchatscreen.com.

The M-CHAT-R/F is a copyrighted instrument, and use of the M-CHAT-R/F must follow these guidelines:

(1) Reprints/reproductions of the M-CHAT-R must include the copyright at the bottom (© 2009 Robins, Fein, & Barton). No modifications can be made to items, instructions, or item order without permission from the authors.

(2) The M-CHAT-R must be used in its entirety. Evidence indicates that any subsets of items do not demonstrate adequate psychometric properties.

(3) Parties interested in reproducing the M-CHAT-R/F in print (e.g., a book or journal article) or electronically for use by others (e.g., as part of digital medical record or other software packages) must contact Diana Robins to request permission (DianaLRobins@gmail.com).

(4) If you are part of a medical practice, and you want to incorporate the first stage M-CHAT-R questions into your own practice's electronic medical record (EMR), you are welcome to do so. However, if you ever want to distribute your EMR page outside of your practice, please contact Diana Robins to request a licensing agreement.

Instructions for Use

The M-CHAT-R can be administered and scored as part of a well-child care visit, and also can be used by specialists or other professionals to assess risk for ASD. The primary goal of the M-CHAT-R is to maximize sensitivity, meaning to detect as many cases of ASD as possible. Therefore, there is a high false positive rate, meaning that not all children who score at risk will be diagnosed with ASD. To address this, we have developed the Follow-Up questions (M-CHAT-R/F). Users should be aware that even with the Follow-Up, a significant number of the children who screen positive on the M-CHAT-R will not be diagnosed with ASD; however, these children are at high risk for other developmental disorders or delays, and therefore, evaluation is warranted for any child who screens positive. The M-CHAT-R can be scored in less than two minutes. Scoring instructions can be downloaded from http://www.mchatscreen.com. Associated documents will be available for download as well.

Scoring Algorithm

For all items except 2, 5, and 12, the response "NO" indicates ASD risk; for items 2, 5, and 12, "YES" indicates ASD risk. The following algorithm maximizes psychometric properties of the M-CHAT-R:

LOW-RISK: **Total Score is 0-2**; if child is younger than 24 months, screen again after second birthday. No further action required unless surveillance indicates risk for ASD.

MEDIUM-RISK: **Total Score is 3-7**; Administer the Follow-Up (second stage of M-CHAT-R/F) to get additional information about at-risk responses. If M-CHAT-R/F score remains at 2 or higher, the child has screened positive. Action required: refer child for diagnostic evaluation and eligibility evaluation for early intervention. If score on Follow-Up is 0-1, child has screened negative. No further action required unless surveillance indicates risk for ASD. Child should be rescreened at future well-child visits.

HIGH-RISK: **Total Score is 8-20**; It is acceptable to bypass the Follow-Up and refer immediately for diagnostic evaluation and eligibility evaluation for early intervention.

© 2009 Diana Robins, Deborah Fein, & Marianne Barton

M-CHAT-R™

Please answer these questions about your child. Keep in mind how your child usually behaves. If you have seen your child do the behavior a few times, but he or she does not usually do it, then please answer **no**. Please circle **yes** or **no** for every question. Thank you very much.

1. If you point at something across the room, does your child look at it? (**FOR EXAMPLE**, if you point at a toy or an animal, does your child look at the toy or animal?)	Yes	No
2. Have you ever wondered if your child might be deaf?	Yes	No
3. Does your child play pretend or make-believe? (**FOR EXAMPLE**, pretend to drink from an empty cup, pretend to talk on a phone, or pretend to feed a doll or stuffed animal?)	Yes	No
4. Does your child like climbing on things? (**FOR EXAMPLE**, furniture, playground equipment, or stairs)	Yes	No
5. Does your child make unusual finger movements near his or her eyes? (**FOR EXAMPLE**, does your child wiggle his or her fingers close to his or her eyes?)	Yes	No
6. Does your child point with one finger to ask for something or to get help? (**FOR EXAMPLE**, pointing to a snack or toy that is out of reach)	Yes	No
7. Does your child point with one finger to show you something interesting? (**FOR EXAMPLE**, pointing to an airplane in the sky or a big truck in the road)	Yes	No
8. Is your child interested in other children? (**FOR EXAMPLE**, does your child watch other children, smile at them, or go to them?)	Yes	No
9. Does your child show you things by bringing them to you or holding them up for you to see – not to get help, but just to share? (**FOR EXAMPLE**, showing you a flower, a stuffed animal, or a toy truck)	Yes	No
10. Does your child respond when you call his or her name? (**FOR EXAMPLE**, does he or she look up, talk or babble, or stop what he or she is doing when you call his or her name?)	Yes	No
11. When you smile at your child, does he or she smile back at you?	Yes	No
12. Does your child get upset by everyday noises? (**FOR EXAMPLE**, does your child scream or cry to noise such as a vacuum cleaner or loud music?)	Yes	No
13. Does your child walk?	Yes	No
14. Does your child look you in the eye when you are talking to him or her, playing with him or her, or dressing him or her?	Yes	No
15. Does your child try to copy what you do? (**FOR EXAMPLE**, wave bye-bye, clap, or make a funny noise when you do)	Yes	No
16. If you turn your head to look at something, does your child look around to see what you are looking at?	Yes	No
17. Does your child try to get you to watch him or her? (**FOR EXAMPLE**, does your child look at you for praise, or say “look” or “watch me”?)	Yes	No
18. Does your child understand when you tell him or her to do something? (**FOR EXAMPLE**, if you don’t point, can your child understand “put the book on the chair” or “bring me the blanket”?)	Yes	No
19. If something new happens, does your child look at your face to see how you feel about it? (**FOR EXAMPLE**, if he or she hears a strange or funny noise, or sees a new toy, will he or she look at your face?)	Yes	No
20. Does your child like movement activities? (**FOR EXAMPLE**, being swung or bounced on your knee)	Yes	No

© 2009 Diana Robins, Deborah Fein, & Marianne Barton

M-CHAT-R Follow-Up (M-CHAT-R/F)™

Permissions for Use

The Modified Checklist for Autism in Toddlers, Revised, with Follow-Up (M-CHAT-R/F; Robins, Fein, & Barton, 2009) is designed to accompany the M-CHAT-R. The M-CHAT-R/F may be downloaded from www.mchatscreen.com.

The M-CHAT-R/F is a copyrighted instrument, and use of this instrument is limited by the authors and copyright holders. The M-CHAT-R and M-CHAT-R/F may be used for clinical, research, and educational purposes. Although we are making the tool available free of charge for these uses, this is copyrighted material and it is not open source. Anyone interested in using the M-CHAT-R/F in any commercial or electronic products must contact Diana L. Robins at DianaLRobins@gmail.com to request permission.

Instructions for Use

The M-CHAT-R/F is designed to be used with the M-CHAT-R; the M-CHAT-R is valid for screening toddlers between 16 and 30 months of age, to assess risk for autism spectrum disorder (ASD). Users should be aware that even with the Follow-Up, a significant number of the children who fail the M-CHAT-R will not be diagnosed with ASD; however, these children are at risk for other developmental disorders or delays, and therefore, follow-up is warranted for any child who screens positive.

Once a parent has completed the M-CHAT-R, score the instrument according to the instructions. If the child screens positive, select the Follow-Up items based on which items the child failed on the M-CHAT-R; only those items that were originally failed need to be administered for a complete interview.

Each page of the interview corresponds to one item from the M-CHAT-R. Follow the flowchart format, asking questions until a PASS or FAIL is scored. Please note that parents may report “maybe” in response to questions during the interview. When a parent reports “maybe,” ask whether most often the answer is “yes” or “no” and continue the interview according to that response. In places where there is room to report an “other” response, the interviewer must use his/her judgment to determine whether it is a passing response or not.

Score the responses to each item on the M-CHAT-R/F Scoring Sheet (which contains the same items as the M-CHAT-R, but Yes/No has been replaced by Pass/Fail). The interview is considered to be a screen positive if the child fails any two items on the Follow-Up. If a child screens positive on the M-CHAT-R/F, it is strongly recommended that the child is referred for early intervention and diagnostic testing as soon as possible. Please note that if the healthcare provider or parent has concerns about ASDs, children should be referred for evaluation regardless of the score on the M-CHAT-R or M-CHAT-R/F.

© 2009 Diana Robins, Deborah Fein, & Marianne Barton

M-CHAT-R Follow-Up™ Scoring Sheet

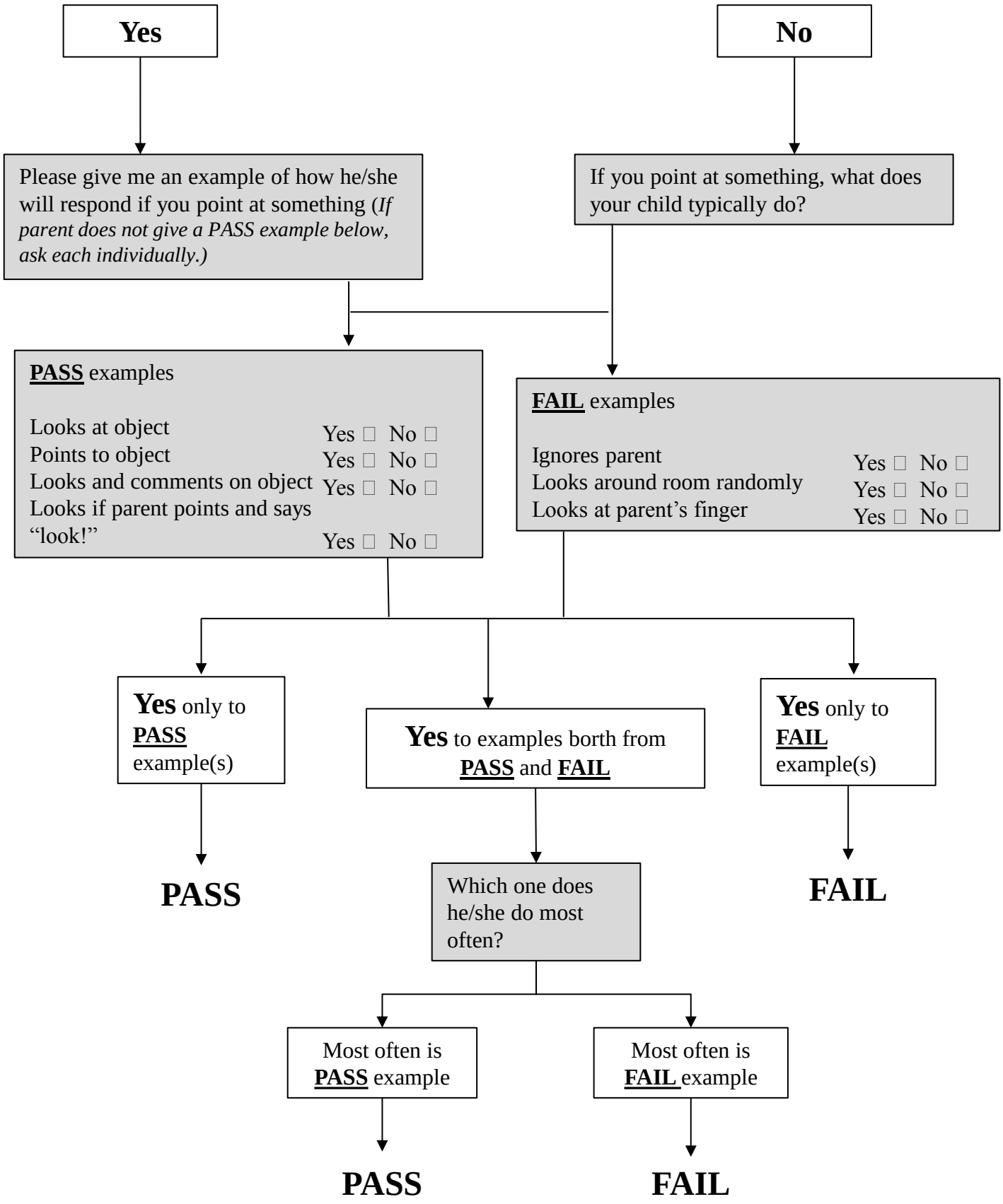
Please note: Yes/No has been replaced with Pass/Fail

1.	If you point at something across the room, does your child look at it? (**FOR EXAMPLE**, if you point at a toy or an animal, does your child look at the toy or animal?)	Pass	Fail
2.	Have you ever wondered if your child might be deaf?	Pass	Fail
3.	Does your child play pretend or make-believe? (**FOR EXAMPLE**, pretend to drink from an empty cup, pretend to talk on a phone, or pretend to feed a doll or stuffed animal)	Pass	Fail
4.	Does your child like climbing on things? (**FOR EXAMPLE**, furniture, playground equipment, or stairs)	Pass	Fail
5.	Does your child make unusual finger movements near his or her eyes? (**FOR EXAMPLE**, does your child wiggle his or her fingers close to his or her eyes?)	Pass	Fail
6.	Does your child point with one finger to ask for something or to get help? (**FOR EXAMPLE**, pointing to a snack or toy that is out of reach)	Pass	Fail
7.	Does your child point with one finger to show you something interesting? (**FOR EXAMPLE**, pointing to an airplane in the sky or a big truck in the road)	Pass	Fail
8.	Is your child interested in other children? (**FOR EXAMPLE**, does your child watch other children, smile at them, or go to them?)	Pass	Fail
9.	Does your child show you things by bringing them to you or holding them up for you to see – not to get help, but just to share? (**FOR EXAMPLE**, showing you a flower, a stuffed animal, or a toy truck)	Pass	Fail
10.	Does your child respond when you call his or her name? (**FOR EXAMPLE**, does he or she look up, talk or babble, or stop what he or she is doing when you call his or her name?)	Pass	Fail
11.	When you smile at your child, does he or she smile back at you?	Pass	Fail
12.	Does your child get upset by everyday noises? (**FOR EXAMPLE**, a vacuum cleaner or loud music)	Pass	Fail
13.	Does your child walk?	Pass	Fail
14.	Does your child look you in the eye when you are talking to him or her, playing with him or her, or dressing him or her?	Pass	Fail
15.	Does your child try to copy what you do? (**FOR EXAMPLE**, wave bye-bye, clap, or make a funny noise when you do)	Pass	Fail
16.	If you turn your head to look at something, does your child look around to see what you are looking at?	Pass	Fail
17.	Does your child try to get you to watch him or her? (**FOR EXAMPLE**, does your child look at you for praise, or say “look” or “watch me”)	Pass	Fail
18.	Does your child understand when you tell him or her to do something? (**FOR EXAMPLE**, if you don’t point, can your child understand “put the book on the chair” or “bring me the blanket”)	Pass	Fail
19.	If something new happens, does your child look at your face to see how you feel about it? (**FOR EXAMPLE**, if he or she hears a strange or funny noise, or sees a new toy, will he or she look at your face?)	Pass	Fail
20.	Does your child like movement activities? (**FOR EXAMPLE**, being swung or bounced on your knee)	Pass	Fail

Total Score: _____

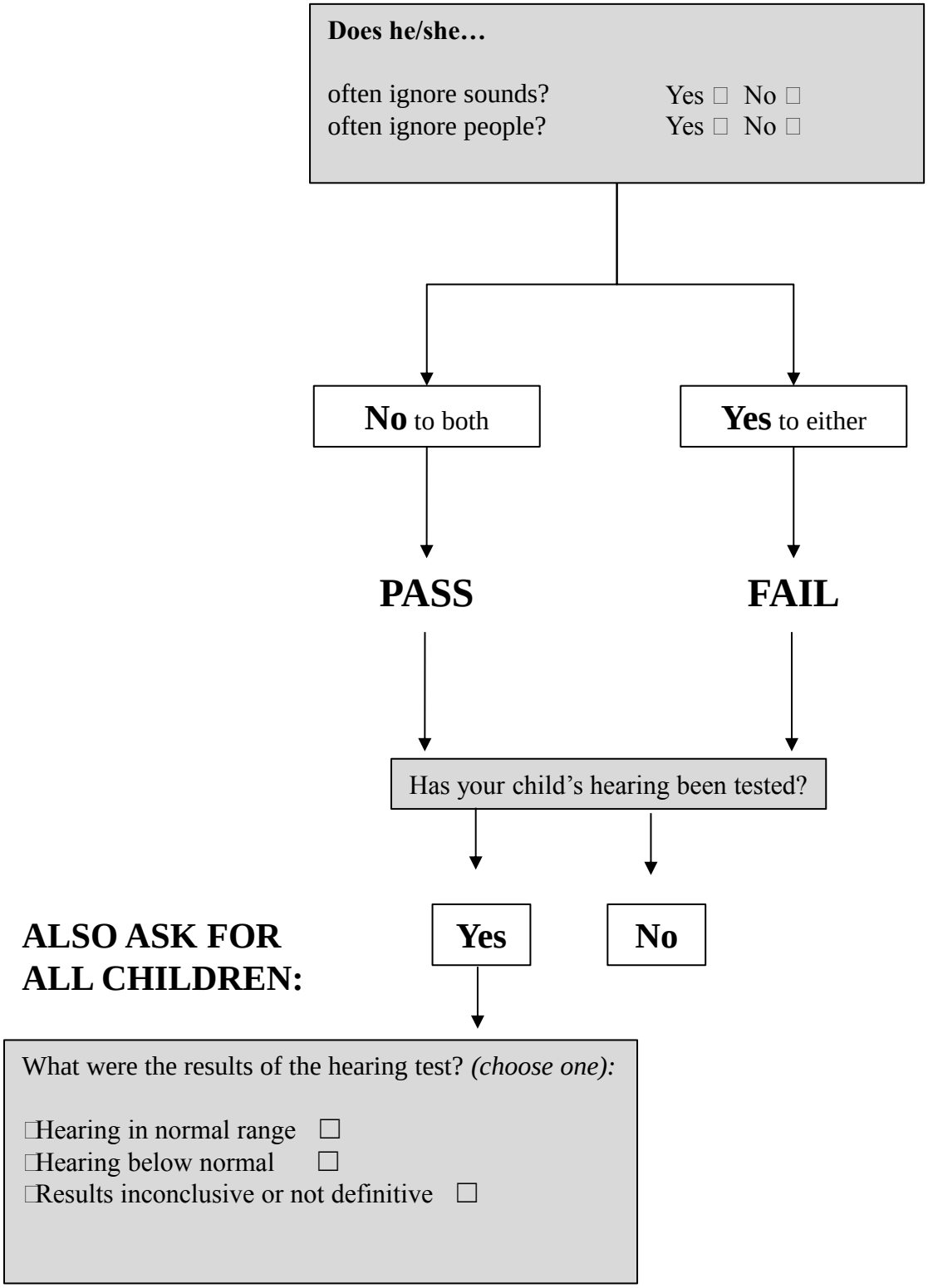
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

1. If you point at something across the room, does _____ look at it?



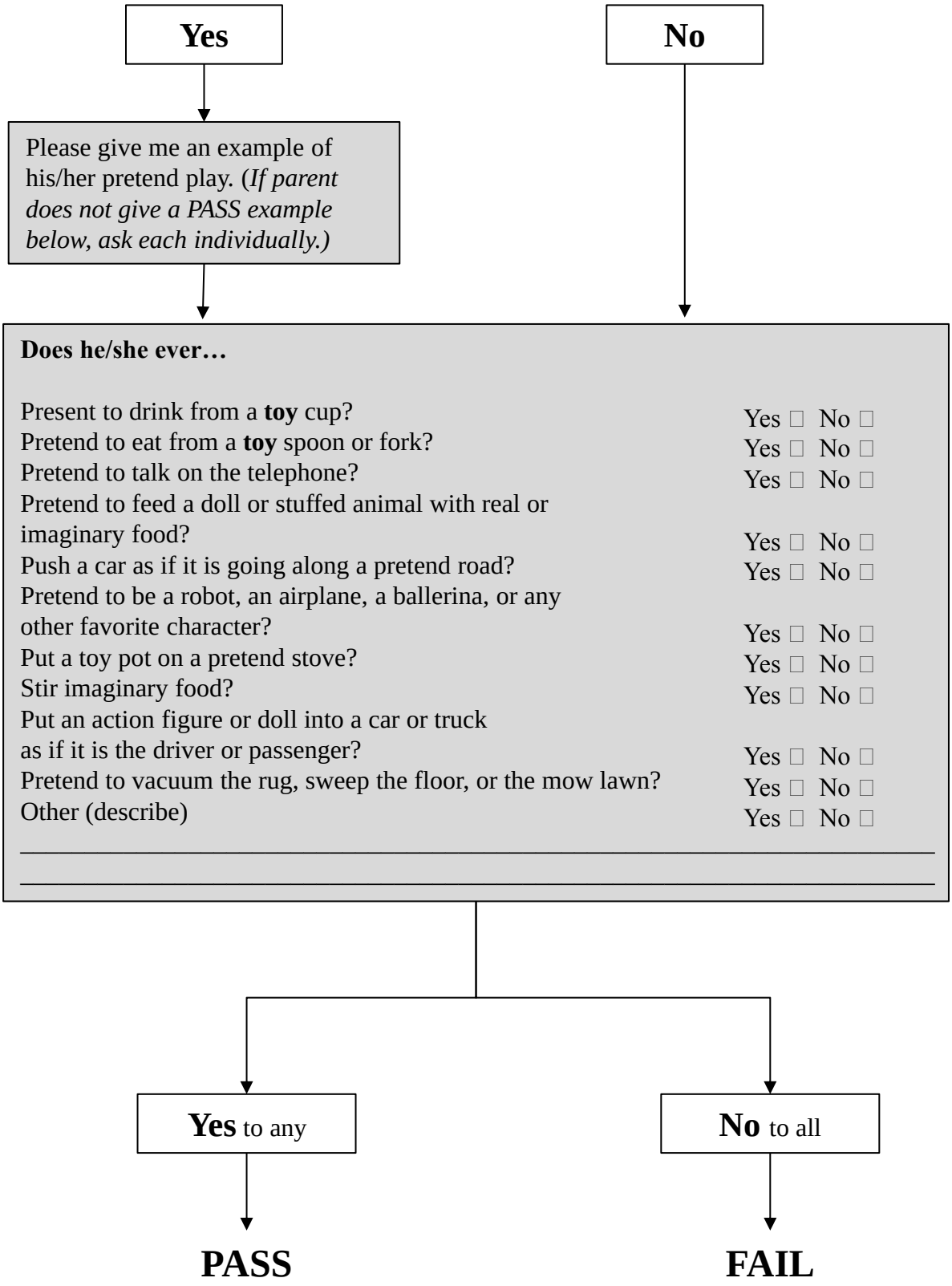
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

2. You reported that you have wondered if you child is deaf. What led you to wonder that?



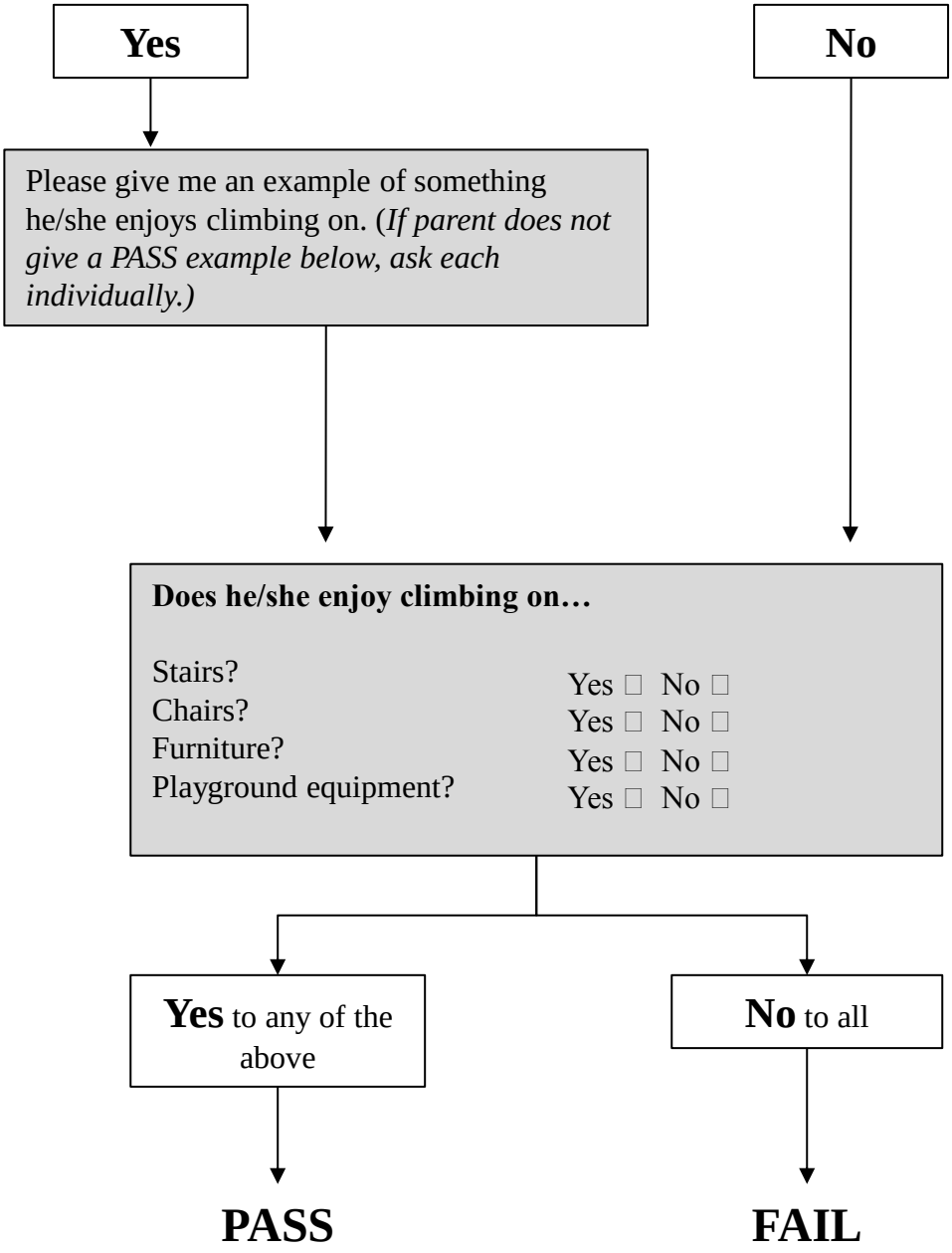
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

3. Does _____ play pretend or make- believe



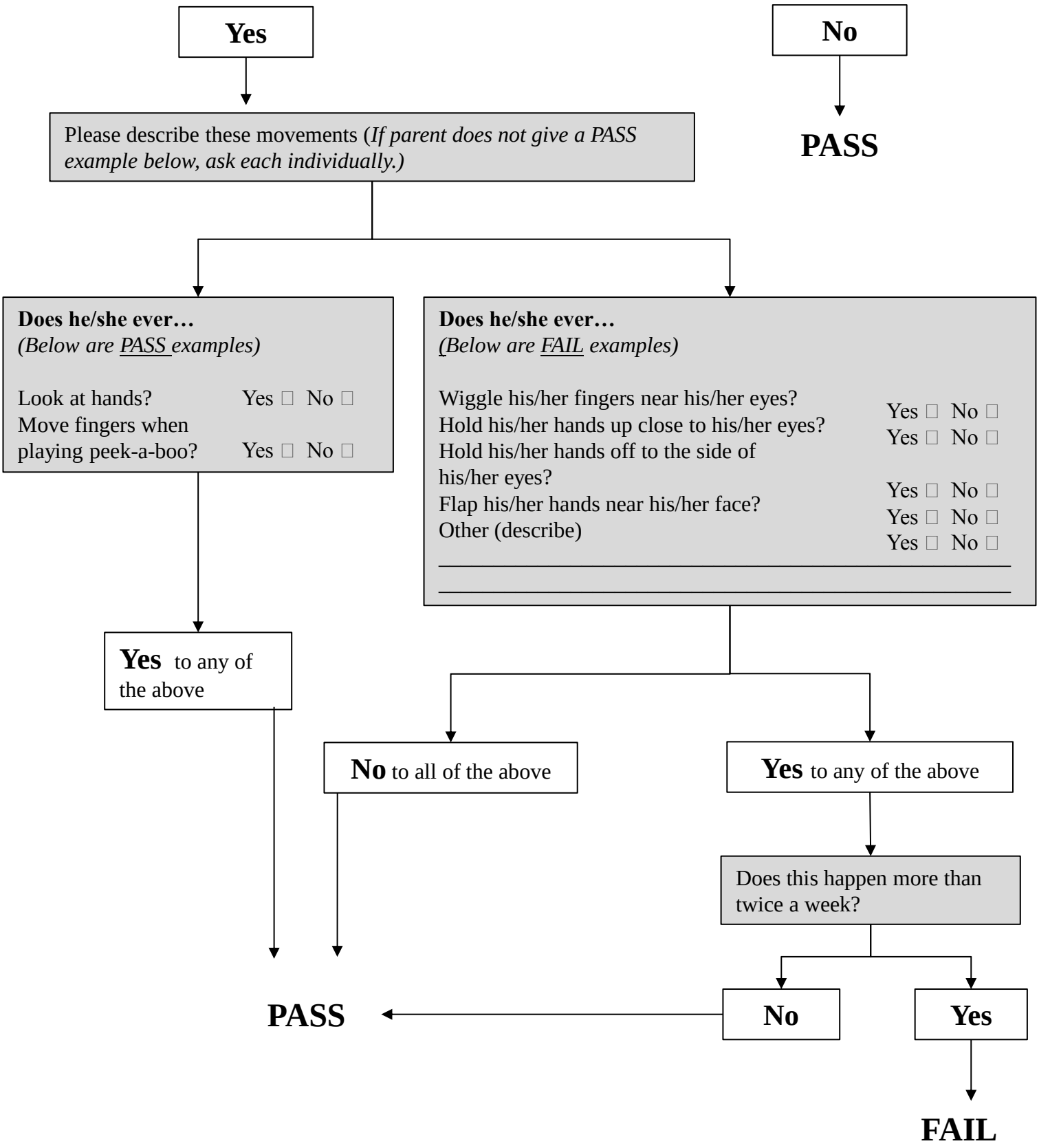
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

4. Does _____ like climbing on things?



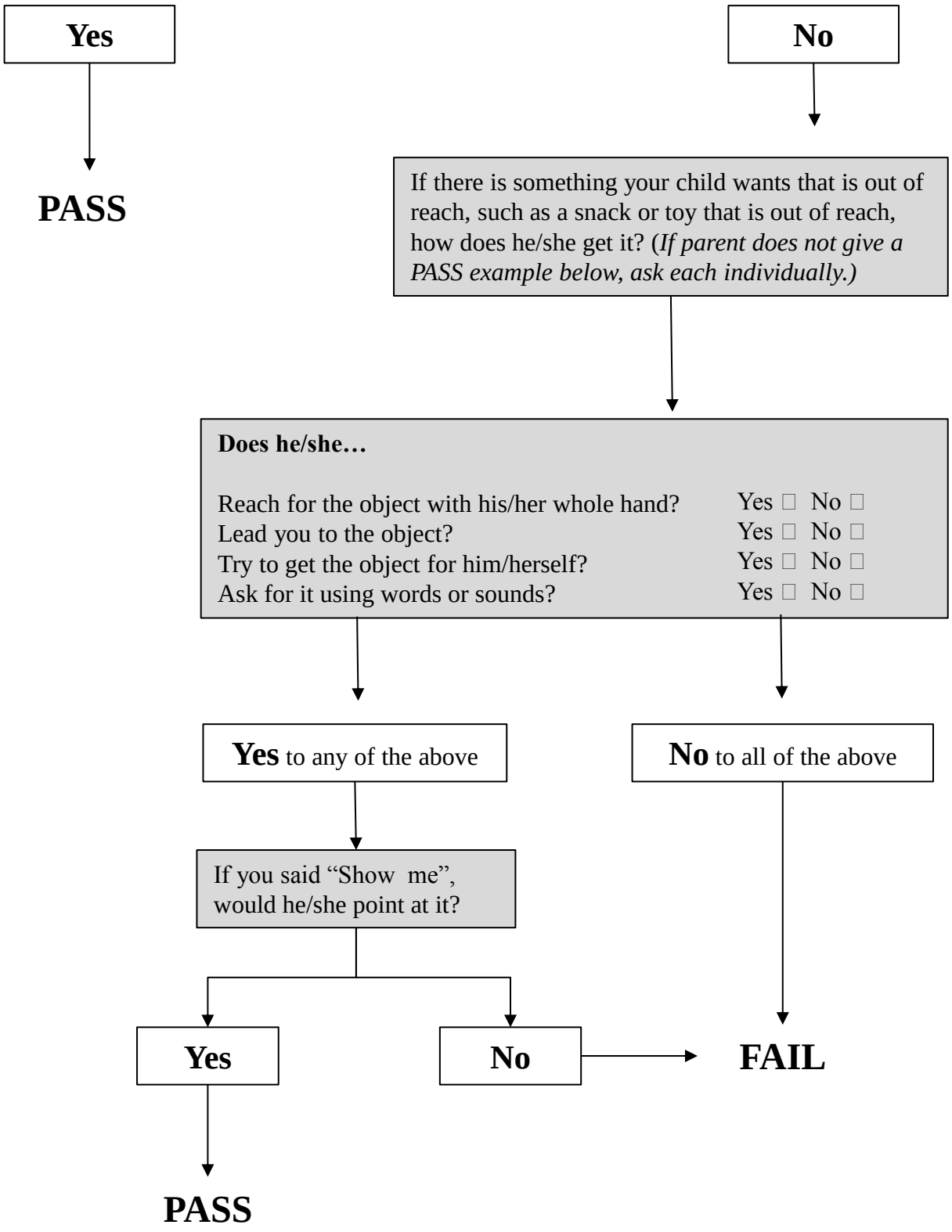
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

5. Does _____ make unusual finger movements near his/her eyes?



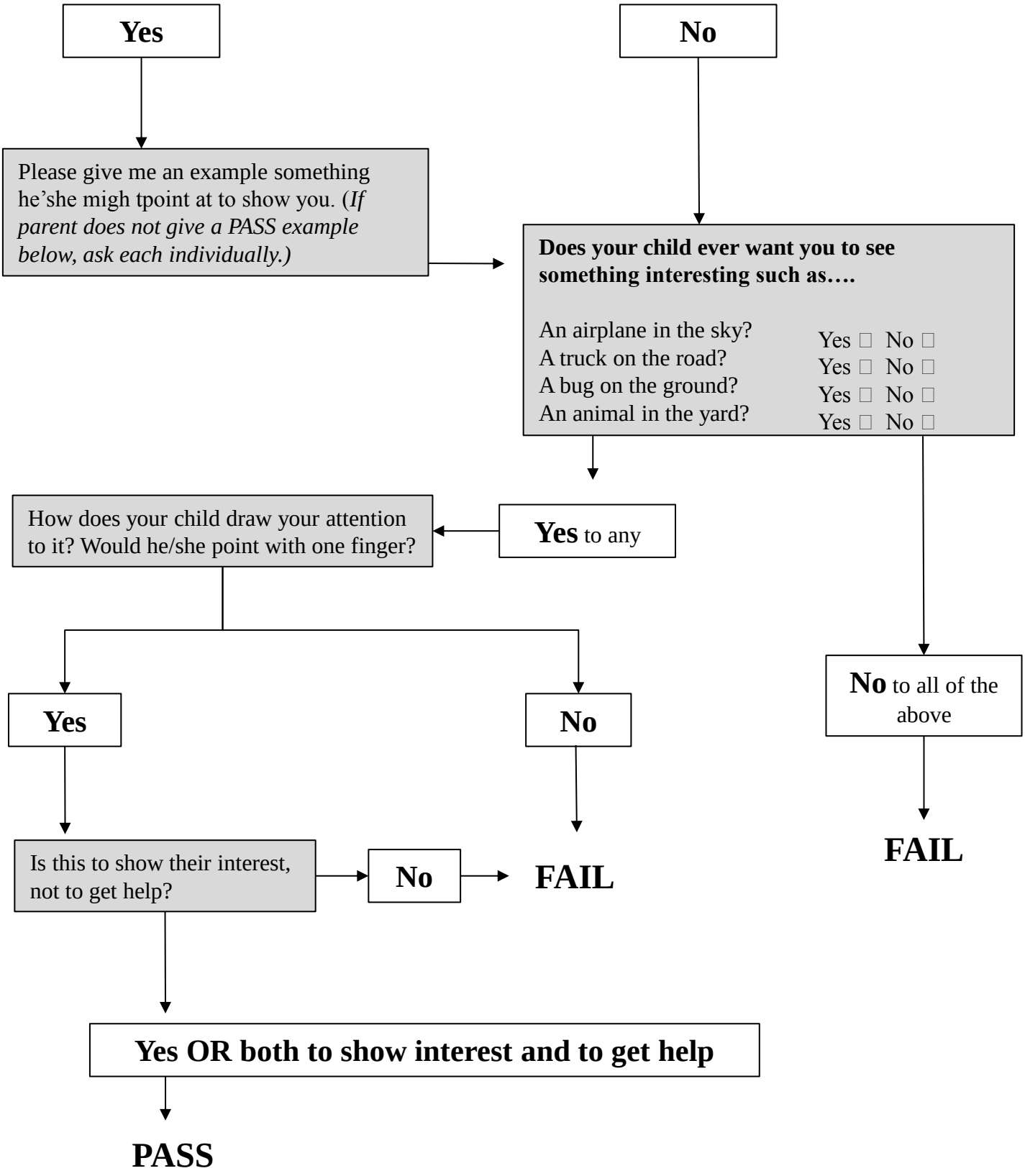
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

6. Does your child point with one finger to ask for something or to get help?



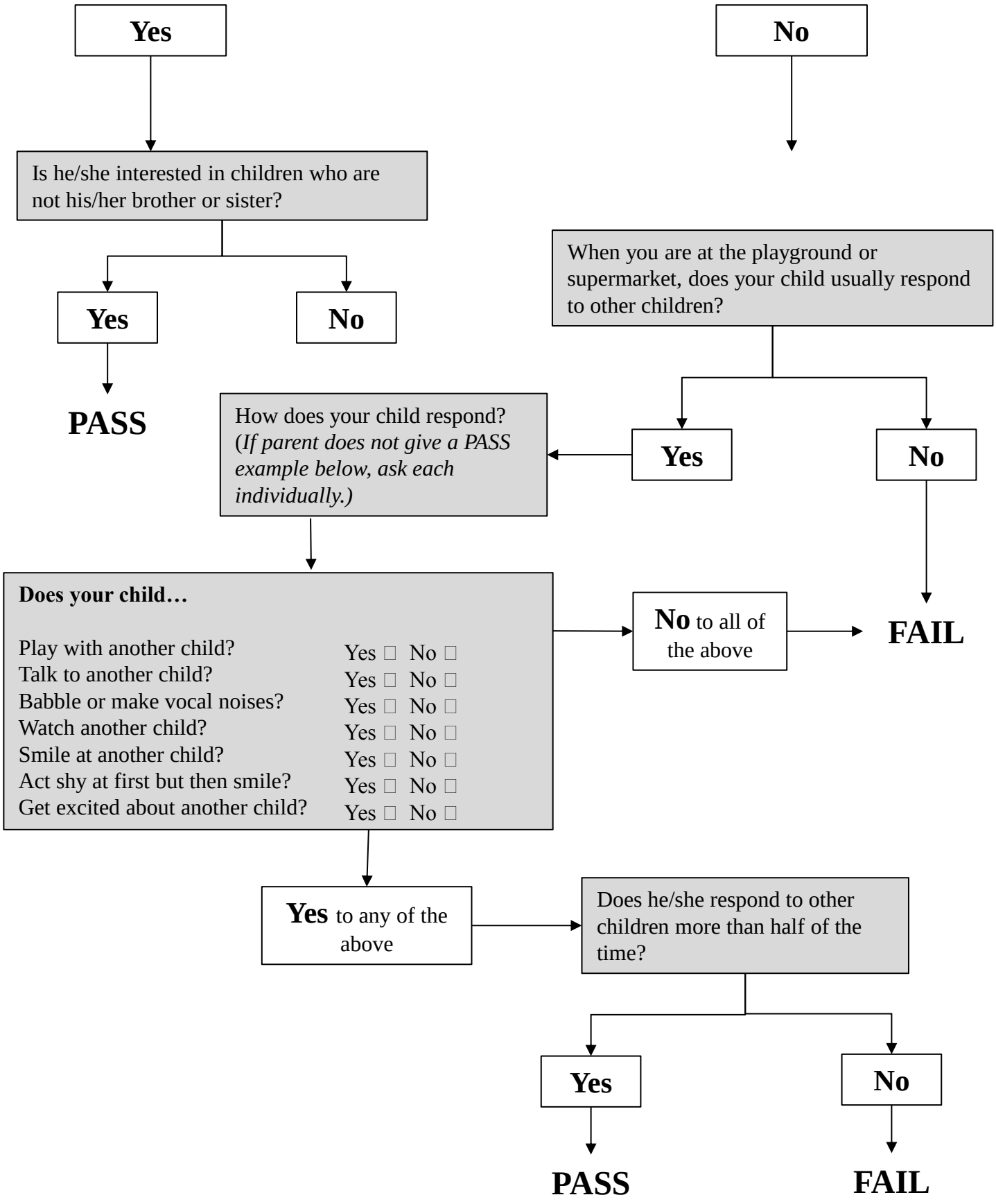
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

7. * **If the interviewer just asked #6, begin here: We just talked about pointing to *ask* for something, ASK ALL → Does your child point with one finger just to show you something interesting?**



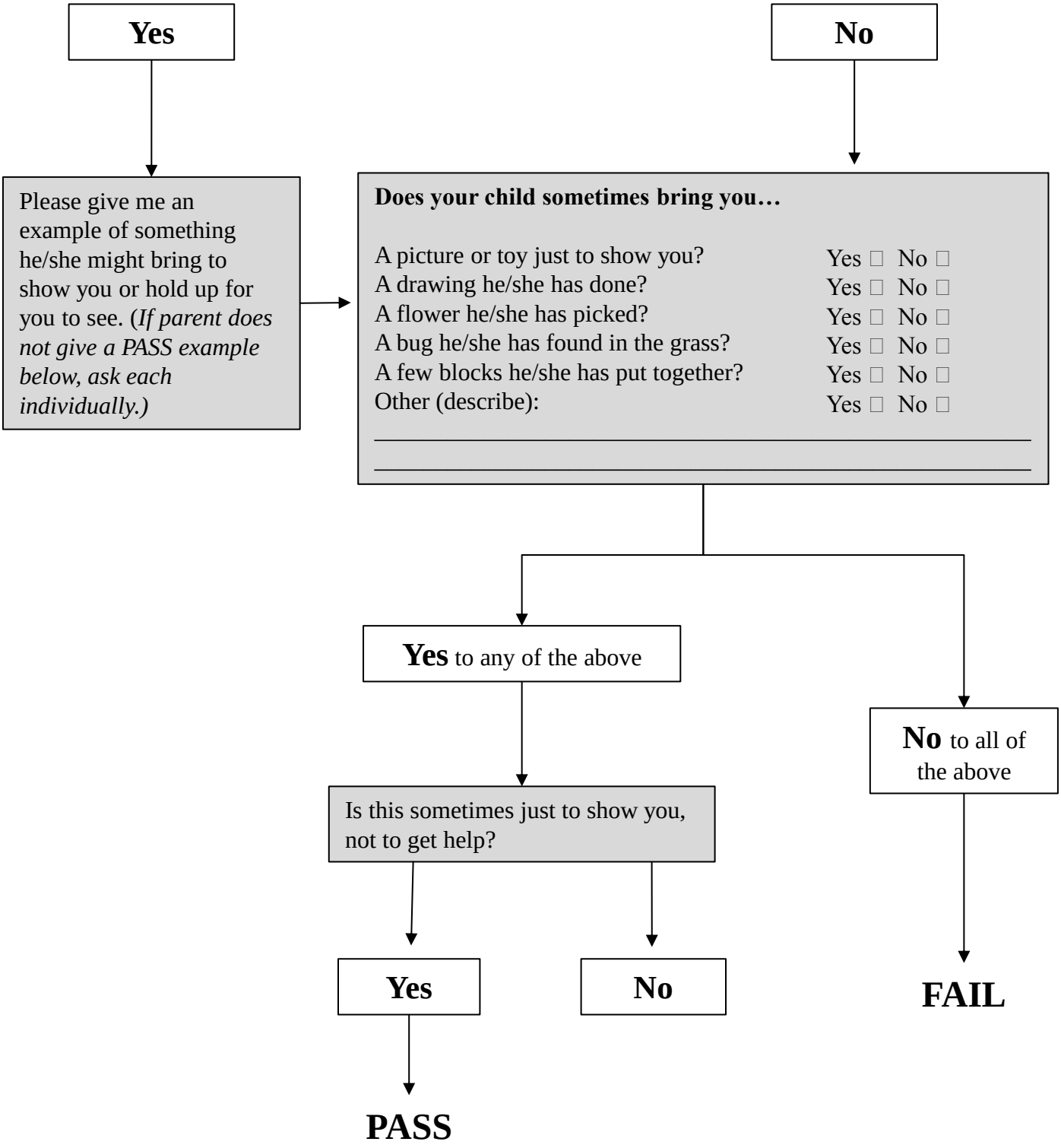
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

8. Is _____ interested in other children?



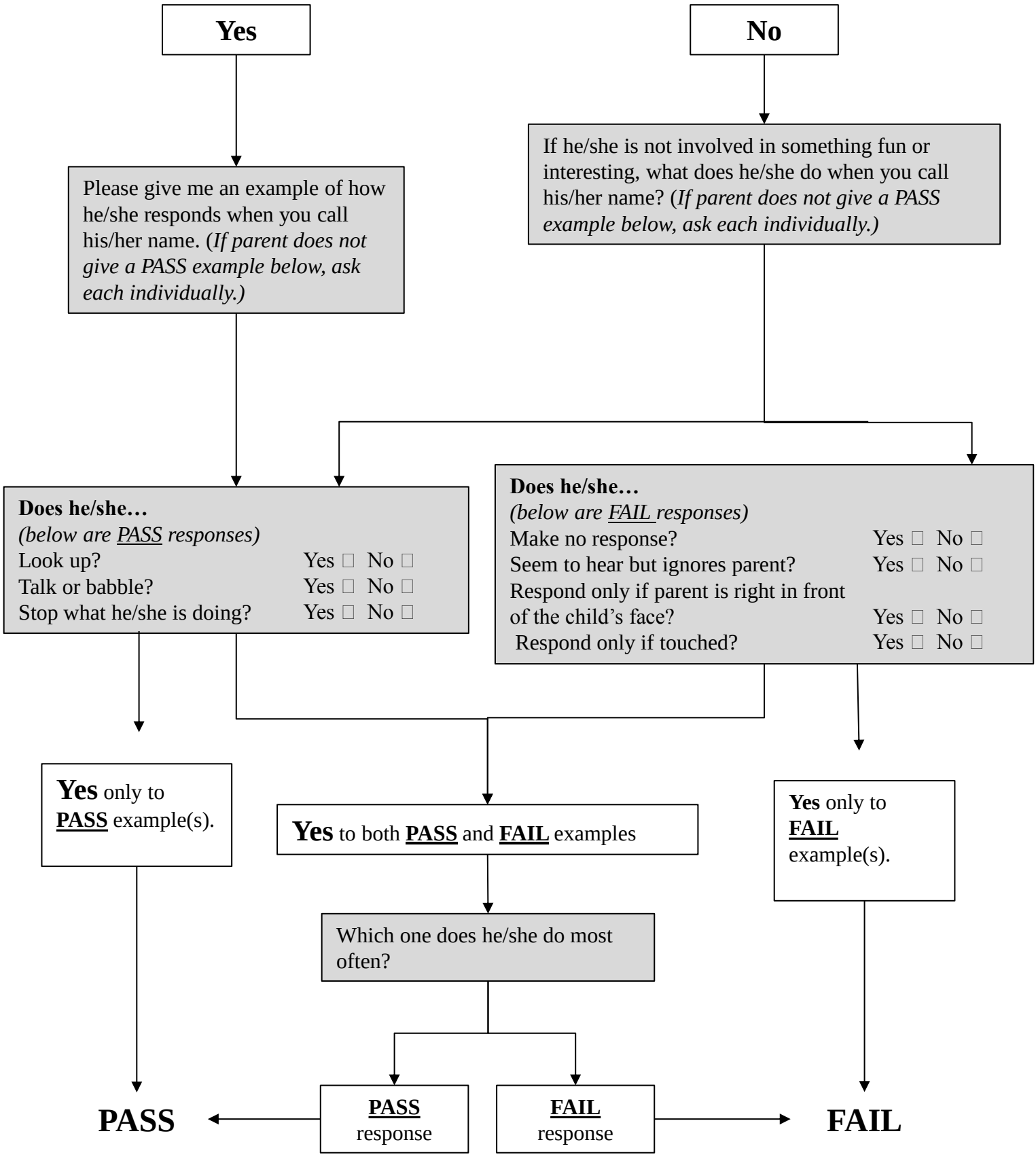
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

9. Does _____ show you things by bringing them to you or holding them up for you to see? Not just to get help, but to share?



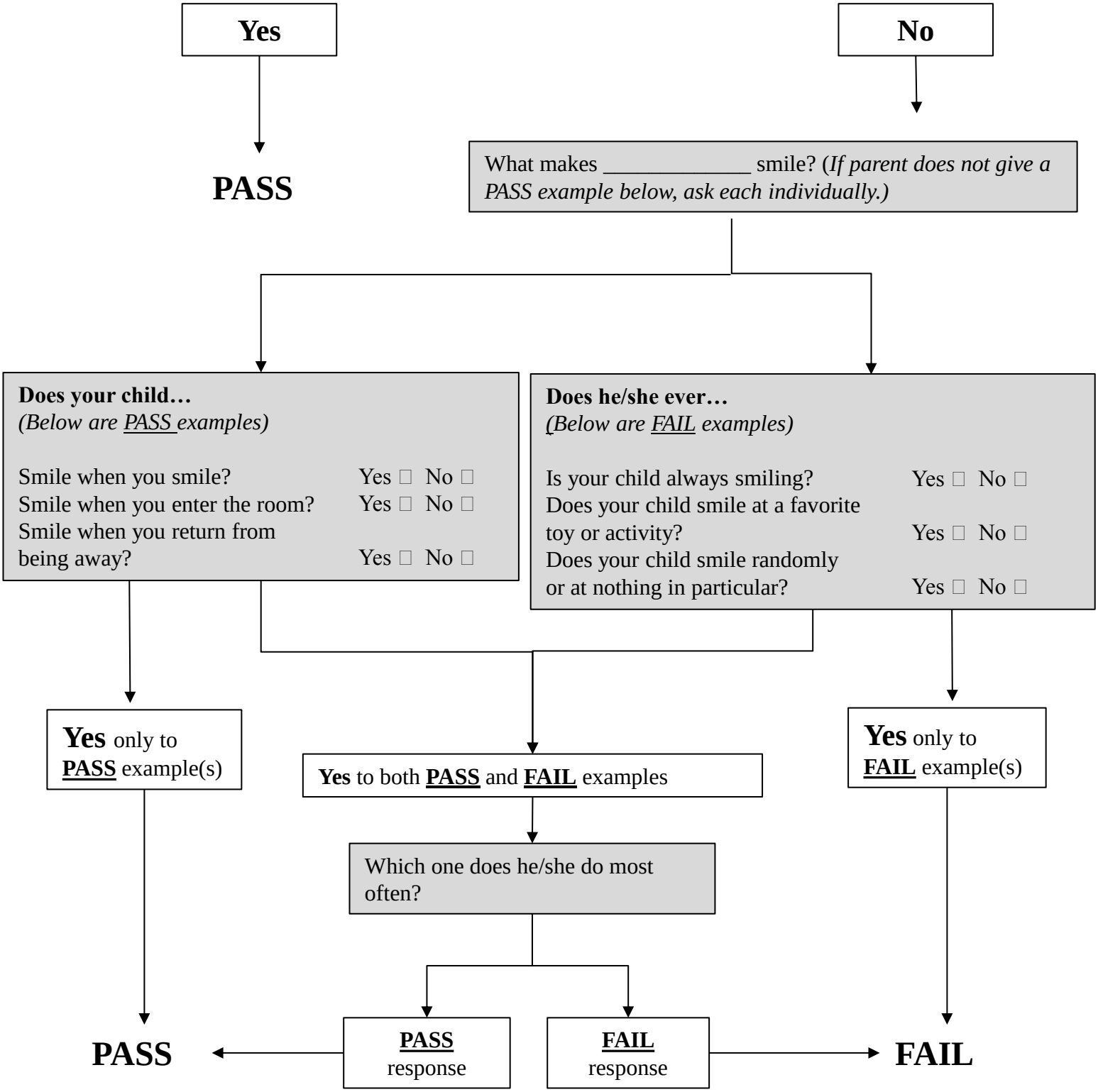
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

10. Does _____ respond when you call his/her name?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton

11. When you smile at _____, does he/she smile back at you?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton

12. Does _____ get upset by everyday noises?

Yes

No

PASS

Does your child have a negative reaction to the sound of…

A washing machine?	Yes ☐ No ☐
Babies crying?	Yes ☐ No ☐
Vacuum cleaner?	Yes ☐ No ☐
Hairdryer?	Yes ☐ No ☐
Traffic?	Yes ☐ No ☐
Babies squealing or screeching?	Yes ☐ No ☐
Loud music?	Yes ☐ No ☐
Telephone/ doorbell ringing?	Yes ☐ No ☐
Noisy places such as a supermarket or restaurant?	☐ Yes ☐ No ☐
Other (describe):	Yes ☐ No ☐

_____

_____

Yes to two or more

How does your child react those noises? (*If parent does not give a PASS example below, ask each individually.)*

Does your child…
(Below are PASS responses)

Calmly cover his/her ears?	Yes ☐ No ☐
Tell you that he/she does not like the noise?	Yes ☐ No ☐

Does your child…
(Below are FAIL responses)

Scream?	Yes ☐ No ☐
Cry?	Yes ☐ No ☐
Cover his/her ears while upset?	Yes ☐ No ☐

Yes only to **PASS** example(s)

Yes to both **PASS** and **FAIL**

Yes only to **FAIL** example(s)

Which one does he/she do most often?

PASS response

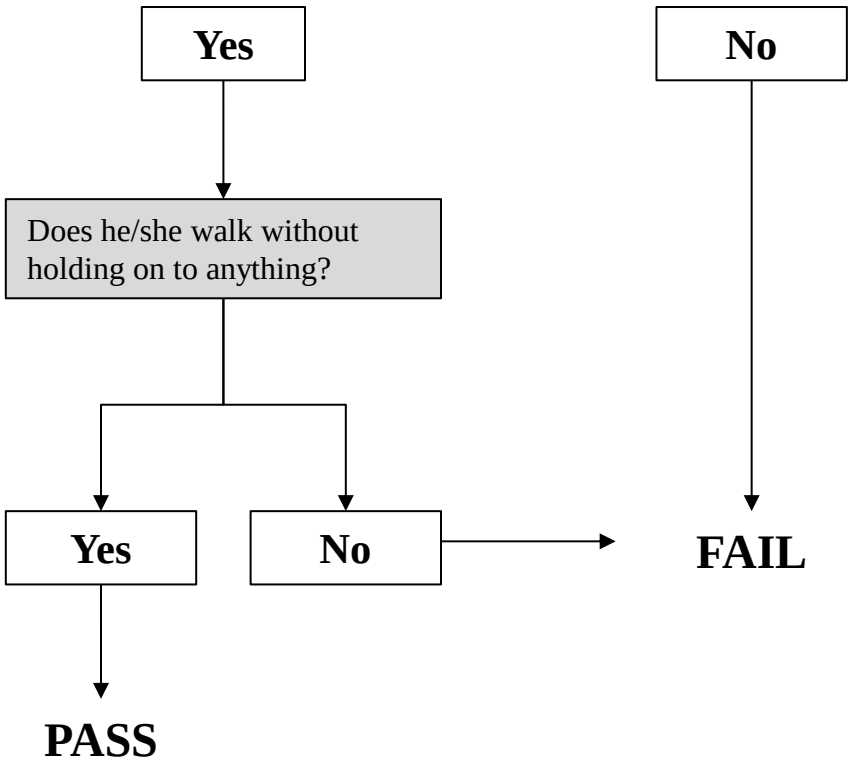
FAIL response

PASS

FAIL

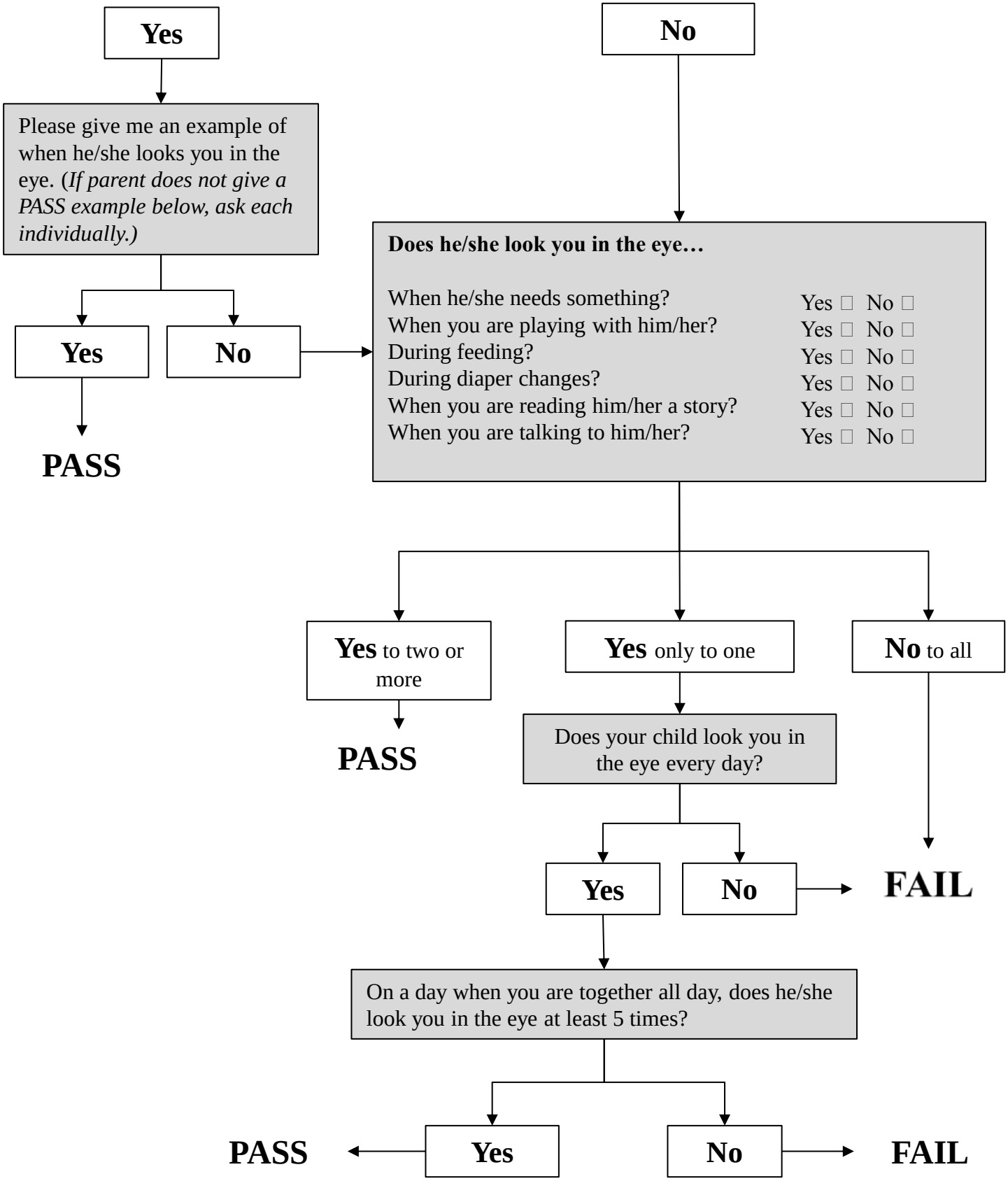
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

13. Does _____ walk?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton

14. Does _____ look you in the eye when you are talking to him/her, playing with him/her, or changing him/her?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton

15. Does _____ try to copy what you do?

Yes

Please give me an example of something he/she would try to copy. (*If parent does not give a PASS example below, ask each individually.)*

No

Does your child try to copy you if you…

Stick out your tongue?	Yes ☐ No ☐
Make a funny sound?	Yes ☐ No ☐
Wave good bye?	Yes ☐ No ☐
Clap your hands?	Yes ☐ No ☐
Put your fingers to your lips to signal "Shhh"?	Yes ☐ No ☐
Blow a kiss?	Yes ☐ No ☐
Other (describe):	Yes ☐ No ☐

_____

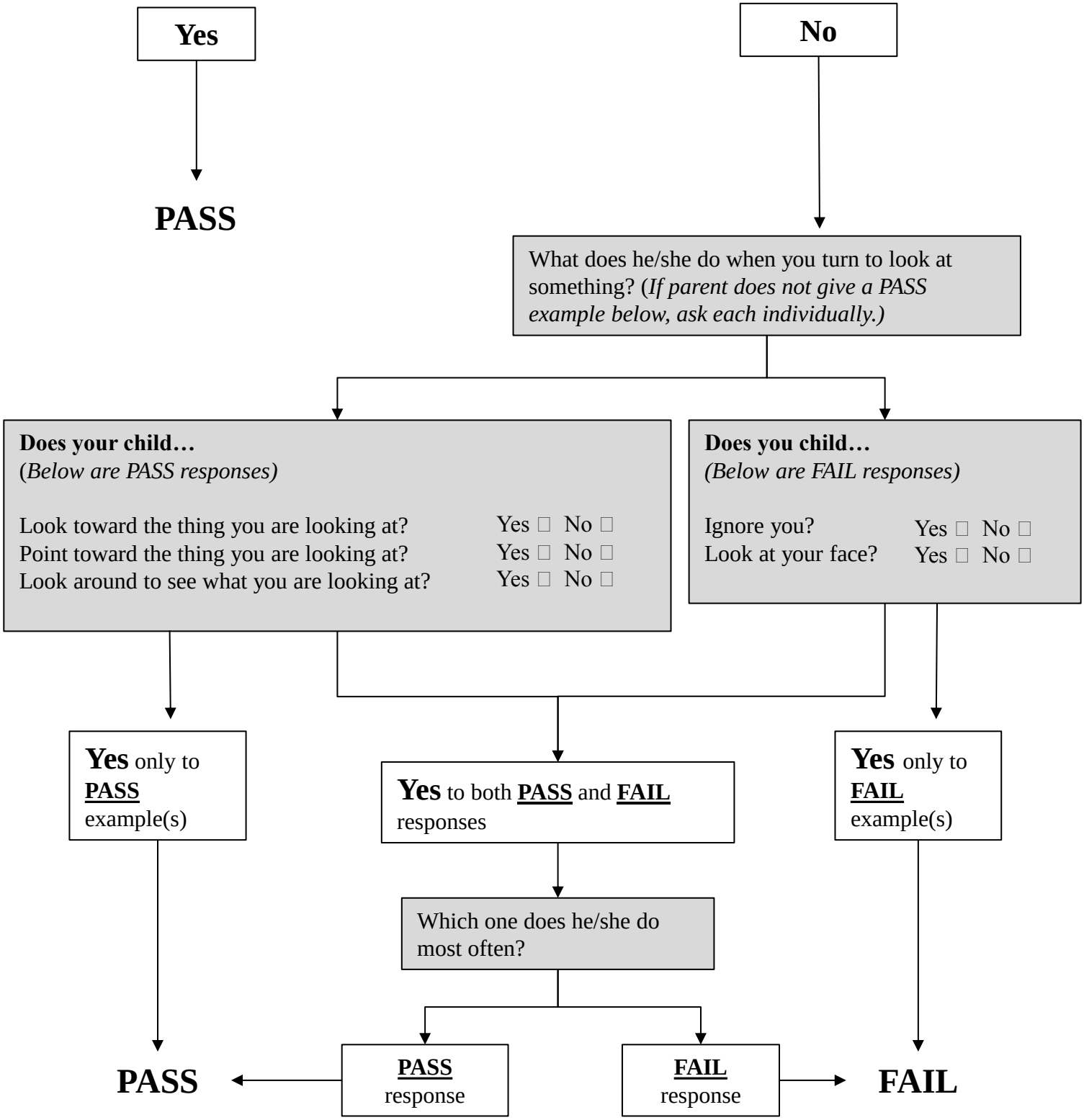
_____

Yes to two or more → **PASS**

Yes to one or none → **FAIL**

© 2009 Diana Robins, Deborah Fein, & Marianne Barton

16. If you turn your head to look at something, does _____ look around to see what you are looking at?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton

17. Does _____ try to get you to watch him/her?

Yes

Please give me an example of how he/she would try to get you to watch him/her. (*If parent does not give a PASS example below, ask each individually.)*

No

Does he/she…

Say "Look!" or "Watch me!"?	Yes ☐	No ☐
Babble or make a noise to get you to watch what he/ she is doing?	Yes ☐	No ☐
Look at you to get praise or comment?	Yes ☐	No ☐
Keep looking to see if you are looking?	Yes ☐	No ☐
Other (describe):	Yes ☐	No ☐

_____

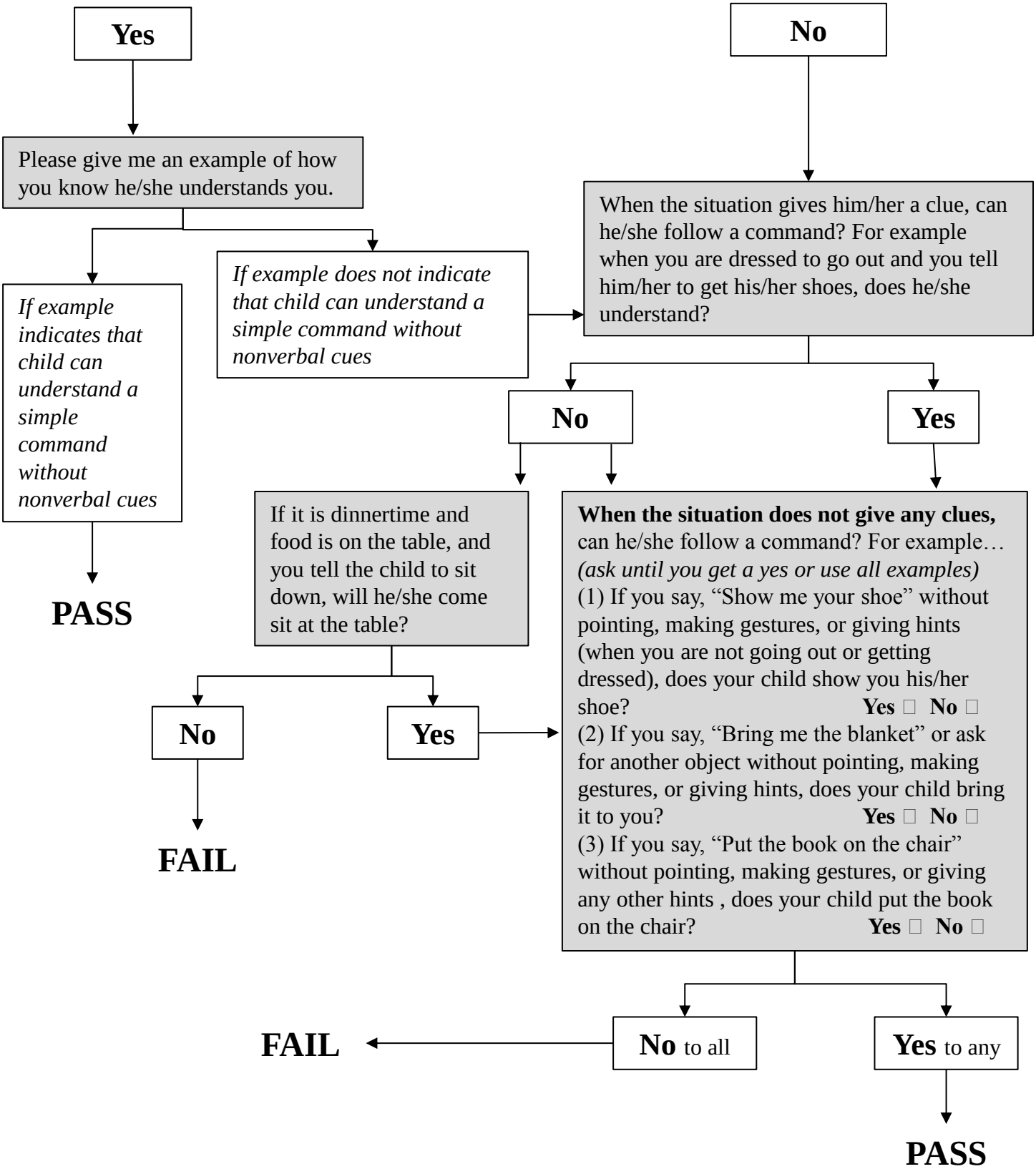
_____

Yes to any → **PASS**

Yes to none → **FAIL**

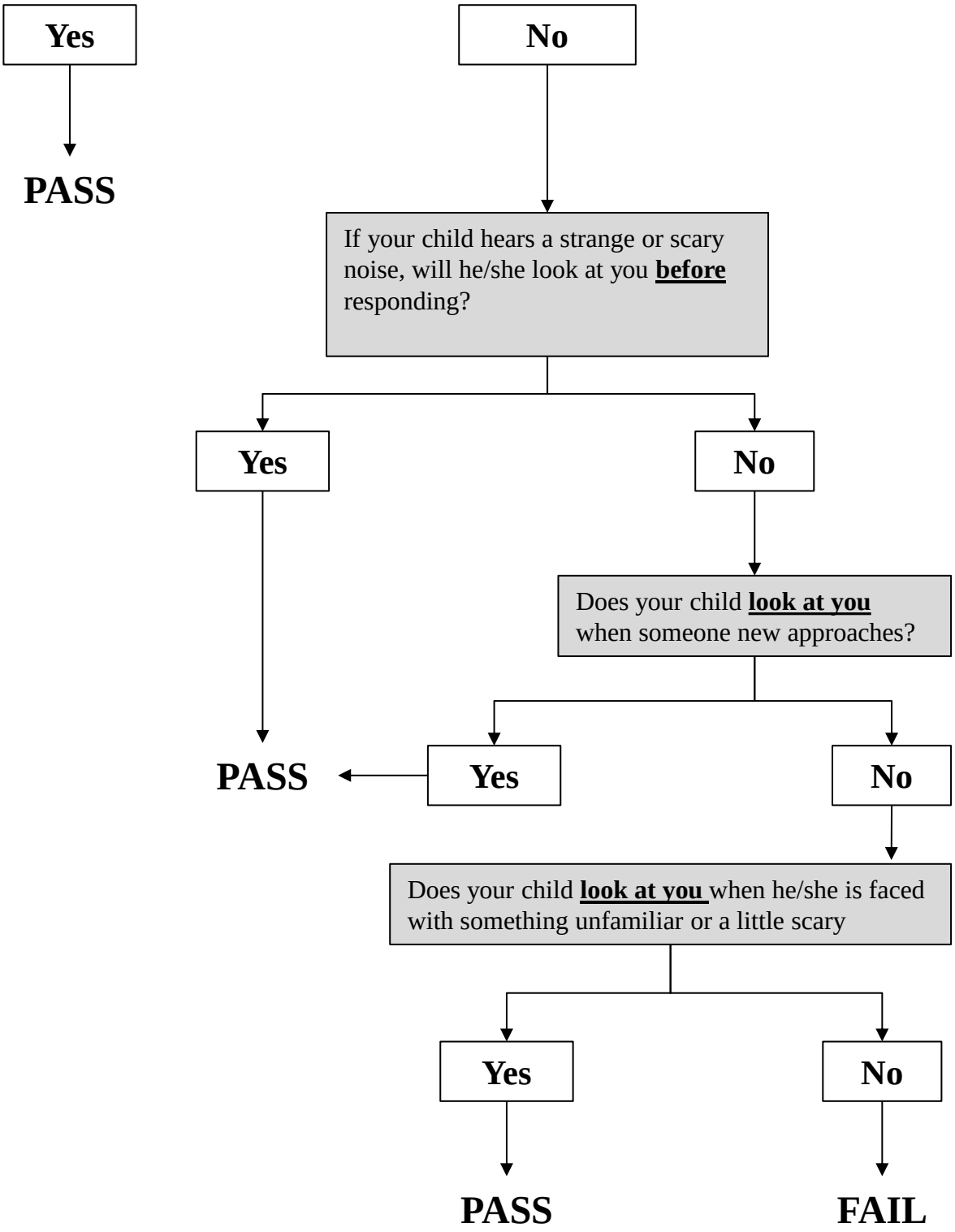
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

18. Does _____ understand when you tell him/her to do something?



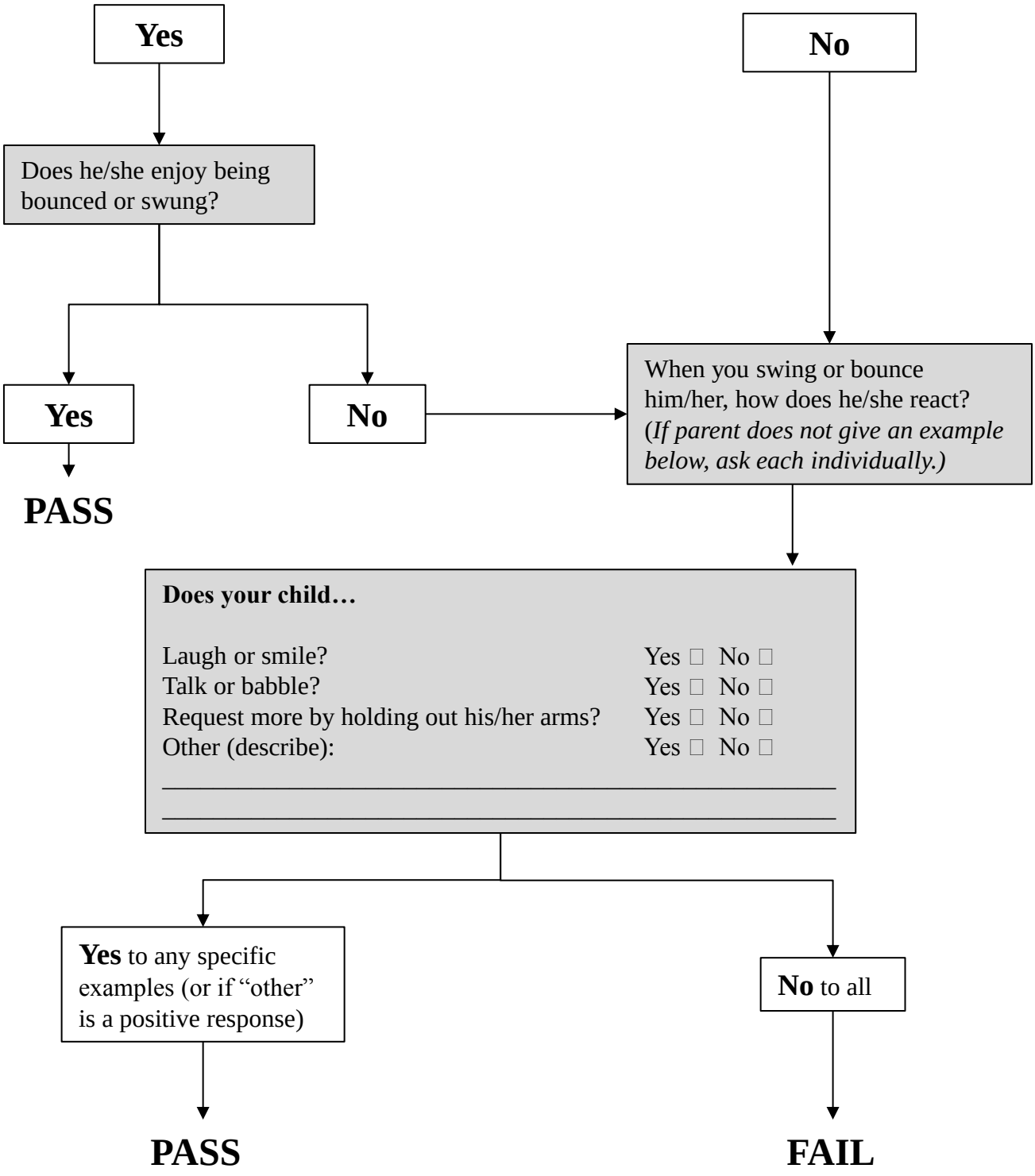
© 2009 Diana Robins, Deborah Fein, & Marianne Barton

19. If something new happens, does _____ look at your face to see how you feel about it?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton

20. Does _____ like movement activities?



© 2009 Diana Robins, Deborah Fein, & Marianne Barton